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Secure product dispenser

Abstract

A secure product dispenser assembly includes a base extending along a base axis between a first end and an opposing second end. The secure product dispenser assembly further includes a top portion configured to adjustably couple to the base and a dispenser portion coupled to the top portion and positioned at least partially over the first end of the base. The base is configured to hold a plurality of units of product and the dispenser portion is configured to rotate between an open position and a closed position. When the dispenser portion is in the open position, the dispenser portion inhibits removal of more than one unit of product from the plurality of units of product held by the base. When the dispenser portion is in the closed position, the dispenser portion is configured to inhibit removal of any of the plurality of units of product held by the base.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This patent application is a divisional patent application of U.S. patent application Ser. No. 17/584,902, filed on Jan. 26, 2022, which claims the priority and benefit of U.S. Provisional Patent Application 63/141,565, filed on Jan. 26, 2021, and is a continuation-in-part of, and claims the priority and benefit of, U.S. patent application Ser. No. 16/840,028, now U.S. Pat. No. 11,363,884, filed on Apr. 3, 2020, which claims the priority and the benefit of U.S. Provisional Patent Application No. 62/828,815, filed on Apr. 3, 2019, and U.S. Provisional Patent Application No. 62/990,090, filed on Mar. 16, 2020. The entire contents of such applications are hereby incorporated by reference.

FIELD OF TECHNOLOGY

(1) The subject matter disclosed herein relates to secure product dispensers, including secure retail displays for products with bottle-necks or similar features (e.g., liquor bottles), or products without bottle necks (e.g., powdered baby formula).

BACKGROUND

(2) In retail stores, many products are stocked openly and freely on shelves, making them visible and accessible to customers. While ideal for allowing the maximum opportunity for a customer to consider a product for purchase, products are also susceptible to theft. To combat this, especially with more expensive products, a store may place those products behind a counter or within a display cabinet that requires the assistance of a store employee to access the products. Other systems exist wherein products are tethered to a shelf and must be released by a store employee for purchase. As a result of cabinets, counters, tethered systems, and the like, customer access to the products is limited and the shopping experience and likelihood of purchase is diminished.

(3) Another method of preventing theft of products is placing more expensive products on higher shelves, leaving less costly products to lower shelves. For example, in a liquor store, expensive liquors are frequently placed on the top shelf, while less expensive liquors are placed on lower shelves. This practice reduces the likelihood that a customer will purchase the more expensive products, diminishing sales revenue. Furthermore, the liquor bottles still remain openly accessible to a potential thief, who may quickly snatch a bottle and depart without paying or “sweep” many bottles off a shelf at once and run out of the store as fast as possible.

(4) The discussion above is merely provided for general background information and is not intended to be used as an aid in determining the scope of the claimed subject matter.

SUMMARY

(5) An embodiment of a secure product dispenser assembly for holding and dispensing products

comprises a base portion and a top portion. The base portion includes a first end and a second end and configured to accept a plurality of units of product. The base portion comprising a first side comprising a first side interface, an opposing second side comprising a second side interface, and a bottom coupled to the first and second sides. A top portion is configured to adjustably couple to the first side interface and the second side interface of the base portion. A dispenser portion is coupled to the top portion and positioned at the first end of the base. A dispensing area is defined by the dispenser portion and the first end of the base. The dispenser portion is configured to rotate between an open position to enable removal of one unit of product from the dispensing area of the base, and a closed position configured to retain and inhibit removal of the one unit of product from the dispensing area of the base. In the open position, the dispenser portion inhibits removal of more than one unit of product. The dispenser portion comprises a plurality of surface features configured to interact with one or more catch elements to enable incremental rotation of the dispenser portion.

(6) The base portion is configured to accept a plurality of units of product and defining a dispensing area at a first end. The base portion further comprises a first side comprising a first side interface, an opposing second side comprising a second side interface, and a bottom side coupled to the first and second sides. The top portion is configured to adjustably couple to first side interface and the second side interface of the base portion. A dispenser portion is coupled to the top portion and positioned above the dispensing area. The dispenser portion is configured to rotate between an open position to enable removal of one unit of product from the dispensing area of the base, and a closed position configured to retain and inhibit removal of the one unit of product from the dispensing area of the base. In the open position, the dispenser portion inhibits removal of more than one unit of product. The dispenser portion further comprises a plurality of surface features configured to interact with one or more catch elements to enable incremental rotation of the dispenser portion.

(7) In an embodiment, the secure product dispenser assembly includes a lock assembly with a lock member. The lock member may be moved to enable the lock assembly to switch between a loading position and a dispensing position. When in the loading position, the lock assembly holds the dispensing portion in the open position such that units of product can be loaded into the secure product assembly from the first end of the base.

(8) In another embodiment, a secure product dispenser assembly comprises a base portion having a first end and a second end, and being configured to hold a plurality of units of product. The base portion comprises a bottom, a first divider coupled to the bottom, and a second divider positioned opposite the first divider and coupled the bottom. A top portion is configured to adjustably couple to the base portion and a dispenser portion is coupled to the top portion and positioned at the first end of the base portion. The dispenser portion is configured to rotate between an open position to enable removal of one unit of product from the first end of the base, and a closed position inhibiting removal of the one unit of product from the first end of the base.

(9) Additional features and advantages of the present disclosure are described in, and will be apparent from, the following Brief Description of the Drawings and Detailed Description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the features of the invention can be understood, a detailed description of the invention may be had by reference to certain exemplary embodiments, some of which are illustrated in the accompanying drawings. It is to be noted, however, that the drawings illustrate only certain exemplary embodiments of this invention and are therefore not to be considered limiting of its scope, for the scope of the invention encompasses other equally effective embodiments. The drawings are not necessarily to scale, emphasis generally being placed upon

illustrating the features of certain exemplary embodiments of the invention. In the drawings, like numerals are used to indicate like parts throughout the various views. Thus, for further understanding of the invention, reference can be made to the following detailed description, read in connection with the drawings.

- (2) FIG. 1A is a perspective view of an exemplary embodiment of a secure product dispenser assembly with products loaded into the assembly and shown in the closed position.
- (3) FIG. 1B is a perspective view of the secure product dispenser assembly of FIG. 1A with products loaded into the assembly and shown in the open position.
- (4) FIG. 2 is a perspective view of the secure product dispenser assembly of FIG. 1A highlighting the features of the first (left) divider.
- (5) FIG. 3 is a perspective view of the secure product dispenser assembly of FIG. 1 without products loaded into the assembly and shown in the closed position.
- (6) FIG. 4 is an exploded view of an exemplary embodiment of the security dispensing handle along with the spring mechanism and gear mechanism.
- (7) FIG. 5 is an underside view of the security dispensing handle of FIG. 5.
- (8) FIG. 6 is a perspective view of multiple secure product dispenser assemblies secured by a locking assembly.
- (9) FIG. 7A is a sectional view of the locking assembly of FIG. 6 shown in the unlocked position.
- (10) FIG. 7B is a sectional view of the locking assembly of FIG. 6 shown in the locked position.
- (11) FIG. 8A is a perspective view of another exemplary embodiment of a secure product dispenser assembly with products loaded into the assembly and shown in the closed position.
- (12) FIG. 8B is a perspective view of the secure product dispenser assembly of FIG. 8A with products loaded into the assembly and shown in the open position.
- (13) FIG. 9 is a perspective view of another embodiment of the secure product dispenser assembly loaded with two product units.
- (14) FIG. 10 is a rear perspective view of the embodiment of the secure product dispenser assembly of FIG. 9.
- (15) FIG. 11 is a front perspective view of an embodiment of a dispensing handle of the secure product dispenser assembly of FIG. 9 in a closed position.
- (16) FIG. 12 is a close-up front perspective view of the embodiment of the secure product dispenser assembly of FIG. 11.
- (17) FIG. 13 is a close-up front perspective view of the embodiment of a dispensing handle of FIG. 11 in a dispensing position or open position.
- (18) FIG. 14 is a top perspective view of an embodiment of the secure product dispenser assembly of FIG. 9.
- (19) FIG. 15 illustrates a plurality of the secure product dispenser assembly of FIG. 9 arranged side-by-side on a display shelf.
- (20) FIG. 16 illustrates a plurality of the secure product dispenser assembly of FIG. 9 arranged side-by-side on a multiple display shelves.
- (21) FIG. 17 illustrates a close up of some of the plurality of the secure product dispenser assemblies from FIGS. 15 and 16.
- (22) FIG. 18 illustrates a right side perspective view of another embodiment of a secure product dispenser assembly.
- (23) FIG. 19 illustrates a top plan view of the embodiment of the secure product dispenser assembly of FIG. 18.
- (24) FIG. 20 illustrates a front plan view of the embodiment of the secure product dispenser assembly of FIG. 18.
- (25) FIG. 21 illustrates a close-up view of a front end portion of the embodiment of the secure product dispenser assembly.
- (26) FIG. 22 illustrates a bottom perspective view of an embodiment of the secure product

dispenser assembly.

(27) FIG. 23 illustrates a right side perspective view of an embodiment of the secure product dispenser assembly.

(28) FIG. 24 illustrates a front perspective view of an embodiment of the secure product dispenser assembly.

(29) FIG. 25 illustrates a right side perspective view of another embodiment of the secure product dispenser assembly.

(30) FIG. 26A illustrates a top perspective view of an embodiment of the secure product dispenser in a dispensing position.

(31) FIG. 26B illustrates a top perspective view of an embodiment of the secure product dispenser in a loading position.

(32) FIG. 27A illustrates a front perspective view of an embodiment of the secure product dispenser in a dispensing position.

(33) FIG. 27B illustrates a top perspective view of an embodiment of the secure product dispenser in a loading position.

(34) FIG. 28 illustrates a front perspective view of a plurality of secure product dispenser assemblies installed on display shelf.

(35) FIG. 29A illustrates a close-up view of a coupling element of an embodiment of the secure product dispenser assembly.

(36) FIG. 29B illustrates a right side perspective view of an embodiment of the secure product dispenser assembly.

(37) FIG. 30A illustrates a top perspective view of an embodiment of a secure product dispenser configured to couple to adjacent secure product dispensers.

(38) FIG. 30B illustrates a close-up view of the embodiment of FIG. 30A.

DESCRIPTION OF THE INVENTION

(39) The following discussion relates to various embodiments of a reusable surface cover with integrated elements. It will be understood that the herein described versions are examples that embody certain inventive concepts as detailed herein. To that end, other variations and modifications will be readily apparent to those of sufficient skill. In addition, certain terms are used throughout this discussion in order to provide a suitable frame of reference with regard to the accompanying drawings. These terms such as “periphery”, “interior”, “exterior”, “front”, “back”, “inner”, “external”, “top”, “bottom”, and the like are not intended to limit these concepts, except where so specifically indicated. With regard to the drawings, their purpose is to depict salient features of the disclosed subject matter and are not specifically provided to scale.

(40) FIG. 1A is a perspective view of an exemplary embodiment of a secure product dispenser assembly **100** with products **10**, **16** loaded into the assembly **100** and shown in the closed position. FIG. 1B is a perspective view of the secure product dispenser assembly **100** of FIG. 1A with products **10**, **16** loaded into the assembly and shown in the open position. FIG. 2 is a perspective view of the secure product dispenser assembly **100** of FIG. 1A highlighting the features of the first (left) divider **110**. FIG. 3 is a perspective view of the secure product dispenser assembly **100** of FIG. 1 without products loaded into the assembly **100** and shown in the closed position.

(41) As can be seen in these figures, the exemplary embodiment of the secure product dispenser assembly **100** allows products (e.g., a first product having a neck **12** and a base **14** and a second product **16**) to be loaded into a retail display for viewing and access by the customer. The exemplary secure product dispenser assembly **100** includes a first (left) divider **110**, a second (right) divider **120**, a base or tray **130** that also includes a base back portion or a tray back portion **132** and a facing **150**, e.g., on which can be mounted a label or price tag. In some exemplary embodiments, a pushing element such as a spring pusher **140** can be used in the tray **130** to push products **10**, **16** forward when a product is dispensed. In other exemplary embodiments, the tray **130** of the secure product dispenser assembly **100** can be tilted such that gravitational forces can

move the products **10**, **16** forward without a pusher **140**.

(42) Since each of the dividers **110**, **120** have substantially the same features, the following discussion and the Figures illustrate those features of the first (left) divider **110**. For example, the exemplary first (left) divider **110** can be injection molded from a clear material and includes a divider top portion **112**, a divider bottom portion **114**, and a divider back portion **116**. As can be seen best in FIG. 2, the divider bottom portion **114** can be formed as part of the tray **130**.

(43) Each of the dividers **110**, **120** are adjustable in height to accommodate products with different heights. For example, the dividers **110**, **120** can be pre-configured for a particular product height or can be adjusted in the store to accommodate a particular product height. As shown in FIG. 6, multiple secure product dispenser assemblies **100** can be arranged side-by-side. And although FIG. 6 shows an example where all of the products are the same and have the same height, it will be understood each of the dividers **110**, **120** of each of the secure product dispenser assemblies **100** can be adjusted separately to accommodate different product heights.

(44) In one example design to provide this height adjustability, the exemplary first (left) divider **110** also includes a divider adjustable height interface **117** that includes a plurality of holes **118** that can mate with a ball-pin **119** located on the divider bottom portion **115**. For example, to set the first (left) divider **110** at the lowest height, the divider adjustable height interface **117** would be inserted into the divider bottom portion **115** until the ball-pin **119** was inserted into the upper or highest hole **118** on the divider adjustable height interface **117**. On the other hand, to set the first (left) divider **110** at the highest height, the divider adjustable height interface **117** would be inserted into the divider bottom portion **115** until the ball-pin **119** was inserted into the lower or lowest hole **118** on the divider adjustable height interface **117**.

(45) The exemplary first (left) divider **110** also includes at least a partial covering configured to extend between a top of the first divider **110** and a top of the second divider **120**. As shown, the partial covering is a divider lip **114** or rail that is positioned above the bottle-neck **12** of the product **10** (along with the similarly positioned divider lip **124** of the second (right) divider **120**) to prevent the product **10** from being lifted vertically out of the exemplary secure product dispenser assembly **100**. As shown, the covering and at least one of the first divider **110** and the second divider **120** may be formed as a single component. In this example, the divider lips **114**, **124** only extend for a portion of the length or depth of the secure product dispenser assembly **100**, leaving space for at least one product **10** to be loaded into the back of the secure product dispenser assembly **100** if necessary. In other exemplary embodiments (e.g., where front or some other loading was used), the divider lips **414**, **424** of the dividers **410**, **420** could extend for the entire the length or depth of the secure product dispenser assembly **400** as shown in FIGS. 8A and 8B.

(46) The exemplary first (left) divider **110** also includes a security dispensing handle mount **113** for receiving and mounting to the security dispensing handle **200**, which, in one exemplary embodiment, is injected molded from an opaque material to include a one-way snap fit connection to the security dispensing handle mounts **113** of both dividers **110**, **120** so that the security dispensing handle **200** cannot be removed from the dividers **110**, **120** without breaking the security dispensing handle **200** or the dividers **110**, **120**.

(47) As seen in FIGS. 1A, 3, and 6, when the security dispensing handle **200** is in the home or closed position, the product **10** cannot be removed from the secure product dispenser assembly **100** the security dispensing handle **200** prevents the product **10** from being pulled forward. The product **10** contacts the inner ring **202** of the security dispensing handle **200** preventing the product **10** from moving forward.

(48) As shown in FIG. 1B, when the security dispensing handle **200** is rotated (e.g., using its gripping element **210**) to the open position (e.g., in clockwise direction), the first product **10** in the row of products can be removed from the secure product dispenser assembly **100**, but the security dispensing handle **200** and the divider lips **114**, **124** prevent any products **16** from being removed from the secure product dispenser assembly **100** while the security dispensing handle **200** returns to

its home or closed position. The second product **16** contacts the outer ring **204** of the security dispensing handle **200** preventing the preventing the next product **16** from moving forward. In this way, the security dispensing handle **200** only permits the removal of a single product **10** at a time from the secure product dispenser assembly **100**.

(49) FIG. 4 is an exploded view of an exemplary embodiment of the security dispensing handle **200** along with the spring mechanism **220** and gear mechanism **230**. FIG. 5 is an underside view of the security dispensing handle **200** of FIG. 5. It will be understood, that the shape and size of the exemplary embodiment of the security dispensing handle **200** shown in FIGS. 1A-6 can be changed depending on the application. For example, as shown in FIGS. 8A and 8B, another exemplary embodiment of the security dispensing handle **500** can have a larger surface area (and can even run the full length of the product) that might allow the addition of graphics or tags to the security dispensing handle **500**.

(50) As will be explained, by controlling (e.g., dampening) the speed at which the security dispensing handle **200** returns from its open position (shown in FIG. 1A) to its home or closed position (shown in FIG. 1B), the secure product dispenser assembly **100** prevents a customer from rapidly grabbing multiple products at the same time or in succession.

(51) In one exemplary embodiment, the spring mechanism **220** is a spring with one end fixedly attached to one of the dividers (e.g., the second (right) divider **120**) and the other end attached to the underside (e.g., a hook) of the security dispensing handle **200**. When the security dispensing handle **200** is in the home or closed position (shown in FIG. 1A), the spring mechanism **220** is in a low tension (or compressed) state. When the security dispensing handle **200** is rotated from the closed position (shown in FIG. 1A) to the open position (shown in FIG. 1B), the spring mechanism **220** is stretched or extended, increasing the tension on the spring as the security dispensing handle **200** is opened and increasing the force in the opposite direction trying to close the security dispensing handle **200**. Without any gear mechanism **230**, the spring mechanism **220** would cause the security dispensing handle **200** to quickly return to the closed position and allow the customer to then quickly open the security dispensing handle **200** to remove another product **16**.

(52) In one exemplary embodiment, the gear mechanism **230** comprises an oil gear mounted to one of the dividers (e.g., the second (right) divider **120**) that interfaces with a teeth rack **232** molded or otherwise mounted on the underside of the security dispensing handle **200**. When the security dispensing handle **200** is rotated from the closed position (shown in FIG. 1A) to the open position (shown in FIG. 1B), the teeth of teeth rack **232** are allowed to roll over the teeth of the gear mechanism **230** without resistance from the gear mechanism **230**. It will be understood that mechanical gearing (e.g., gearing molded as part of the security dispensing handle **200** using coupling of gears with different diameters interacting with each other) could also be used. In one embodiment, one-way teeth can be used in the teeth rack **232** to avoid rotating the gear mechanism **230** when rotating from the closed position to the open position. When the security dispensing handle **200** is released from the open position and, based on force of the spring mechanism **220**, tries to rotate back to the home or closed position, the teeth rack **232** engages with the gear mechanism **230** that provides resistance to and dampens the speed at which the security dispensing handle **200** rotates back to the closed position, delaying that process to take, e.g., several seconds (e.g., 2, 5, 10, 20, 30, 40, 50, 60 seconds, etc. depending on the application). This slow return from open to the home or closed position prevents rapid grabbing of multiple products **10**, **16**. The security dispensing handle **200** also forces the customer to use two hands (one to operate the security dispensing handle **200** and one to grab the product **10**, **16**), which can prevent rapid grabbing of multiple products **10**, **16**.

(53) FIG. 6 is a perspective view of multiple secure product dispenser assemblies **100** secured by a locking assembly **300**. FIG. 7A is a sectional view of the locking assembly **300** of FIG. 6 shown in the unlocked position. FIG. 7B is a sectional view of the locking assembly **300** of FIG. 6 shown in the locked position.

(54) In one exemplary embodiment, in order to prevent the unauthorized removal of the secure product dispenser assemblies **100** from the shelf **340** (e.g., gondola shelf) on which it sits, a locking assembly **300** comprising a first locking rail **310** and a second locking rail **320** is employed. In one embodiment, the locking rails **310**, **320** are sheet metal parts. As will be explained, in one exemplary embodiment, the first locking rail **310** is fastened to the shelf **340**, the second locking rail **320**, is interlocked with the first locking rail **310** and the secure product dispenser assemblies **100**, and the two locking rails **310**, **320** are locked together by one or more locks **350** (e.g., cam locks) effectively locking the secure product dispenser assemblies **100** to the shelf **340** and preventing removal.

(55) In one exemplary embodiment, the secure product dispenser assemblies **100** include a hook **330** molded into, e.g., the tray **130**. One end of the first locking rail **310** sits flat on and is fastened (e.g., with hardware) to the top of the shelf **340** beneath the tray and the shelf **340**. This first locking rail **310** then wraps around and below the front part of the shelf **340**. The second end of the locking rail **310** can include slots **312** that will be used to interface with and provide a hinge for the second locking rail **320**. One end of the second locking rail **320** includes a hook **324** that is shaped to mate and lock with the hook **330** formed as part of the secure product dispenser assemblies **100** in the locked position. The second end of the second locking rail **320** includes teeth **322** that can be inserted through the slots **312** of the first locking rail **310**, interlocking the two locking rails **310**, **320** together and forming a hinge connection.

(56) As shown in FIG. 7A, in the unlocked position, the second locking rail **320** can be swung down below the shelf **340** and out of the way of the secure product dispenser assemblies **100**, allowing the secure product dispenser assemblies **100** to be removed from the shelf **340**. As shown in FIG. 7B, in the locked position, the hook **324** of the second locking rail **320** is secured onto the hook **330** formed as part of the secure product dispenser assemblies **100**. In this locked position, the secure product dispenser assemblies **100** are prevented from being removed from the shelf by the second locking rail **320**, which can be locked to the first locking rail **310** using the one or more locks **350**, preventing the hooks **324**, **330** from being separated. To allow removal of the secure product dispenser assemblies **100**, the cam locks **350** need to be unlocked and the hook **324** of the second locking rail **320** is removed from the hook **330** of the secure product dispenser assemblies **100**. In another exemplary embodiment, the secure product dispenser assemblies **100** could be secured together by dovetailing and tying their trays together so a thief could not remove just a single secure product dispenser assembly **100**. Alternatively, another locking rail system could be incorporated as part of the trays and not to the shelf.

(57) Another embodiment of the secure product dispenser assembly **600** is shown in FIGS. 9-17. In the embodiments shown, the components of the secure product dispenser assembly **600** are comprised of a transparent material. One or more components of the secured product dispenser assembly **600** may be formed as a single unit using injection molding or other manufacturing techniques. As shown, the secure product dispenser assembly **600** generally comprises a base portion **605** and a top portion **670**. The base portion **605** includes a first (left) divider **610**, a second (right) divider **620**, a tray **630** with a tray back portion **640** and a dispensing area **650** at least partially defined by a facing **651** at a first end of the base portion **605**. The facing **651** may be configured or otherwise sized to allow a label, tag, or other indicator may be affixed thereto. An extension **660** may project from a bottom surface **631** of one end of the tray **630** to tilt the tray **630** to enable gravitational forces to move the products **10a**, **16a** from one end of the tray **630** to the other as products **10a**, **16a** are removed from the secured product dispenser assembly **600**. In another embodiment, a spring-loaded pusher is configured to advance additional units of product towards the dispensing area **650** to replace units of product being dispensed to customers.

(58) As shown in FIGS. 9 and 14-17, the extension **660** is configured to lift a back end **601** of the tray **630** relative to the front end **602** or first end of the tray **630** such that the tray **630** is positioned at an angle α relative to a display surface **604** (FIGS. 15-17). The tray **630** may further comprise a

top tray surface **632** and one or more friction reduction elements **634**. As shown in FIG. **10**, the friction reduction elements **634** are raised rails configured to elevate the products **10a**, **16a** from the top tray surface **632** thereby decreasing the surface friction between the top tray surface **632** and the products **10a**, **16a**. The decreased surface friction allows the products **10a**, **16a** to move more freely along the tray **630** when being dispensed. In another embodiment, the friction reduction elements may comprise one or more grooves in the top tray surface **632** that are configured to reduce the surface friction between the top tray surface **632** and the products **10a**, **16a**.

(59) The first and second dividers **610**, **620** may have substantially the same features as the first and second dividers of other embodiments of the secure product dispenser assembly **100**, **400**. In an embodiment, the first and second dividers **610**, **620** and the tray **630** are formed as a single component. In another embodiment, the first and second dividers **610**, **620** may be configured to adjust in height in order to accommodate products of varying heights. The height of the first and second dividers **610**, **620** may be adjusted according to any of the embodiments previously discussed. Multiple secure product dispenser assemblies **600** may be arranged side-by-side on a display surface **604** or shelf in order to securely display and dispense a variety of products. Referring to FIGS. **15-17**, multiple secure product dispenser assemblies **600** are arranged side-by-side on a display shelf **604** such as would be used in a grocery store or pharmacy.

(60) A covering is configured to extend at least partially between the first and second dividers **610**, **620**. As shown, a top portion **670** is configured to extend between and couple to the first and second dividers **610**, **620** and cover the tray **630**, however in other embodiments the covering may be similar to the lip **124** previously discussed. As shown, the top portion **670** does not extend along the entire length of the first and second dividers **610**, **620**, and in an exemplary embodiment, the top portion **670** covers at least the front end **602** of the tray **630** to prevent products **10a**, **16a** from being vertically removed from the secure product dispenser assembly **600**. In an embodiment, the top portion **670** may be configured to be disengaged from the first and second dividers **610**, **620** in order to assist in the loading of product **10a**, **16a**. In an embodiment the top portion **670** may be configured to be locked into engagement with the first and second **610**, **620** dividers to prevent disengagement.

(61) Referring to FIGS. **11-12**, and **14** the top portion **670** further comprises a security dispenser handle coupler **672** that is configured to receive and pivotally couple a security dispensing handle **680**. In an embodiment, the security dispensing handle **680** is injection molded from a transparent or opaque material to include a one-way snap-fit connection with the top portion **670** so that the security dispensing handle **680** cannot be removed from the top portion **670**. In another embodiment the security dispenser handle coupler **672** may comprise a separate fastener. As shown, the security dispensing handle **680** comprises a top surface **682** defining a plurality of surface features **683**, such as teeth. The surface features **683** are configured to engage one or more catch elements (not shown) positioned on the top portion **670**. The security dispensing handle **680** further comprises a lip **684**. As shown in FIGS. **11-13**, the security dispensing handle **680** comprises a substantially circular shape and lip **684** extends partially around the circumference of the security dispensing handle **680**. In a preferred embodiment, the lip **684** extends around approximately half of the circumference of the security dispensing handle **680**.

(62) Referring still to FIGS. **11-12**, the secure product dispenser assembly **600** is depicted in the closed position such that the top portion **670**, the first and second dividers **610**, **620**, the lip **684**, and the facing **651** all act to prevent removal of the products **10a**, **16a** from the first end **602** of the tray **630** of the secure product dispenser assembly **600**. In the closed position, the products **10a**, **16a** may be confined within the secure product dispenser assembly **600**. As shown, the lip **684** is configured to contact and otherwise aid in retaining the top portion of the product **10a**. Referring specifically to A4, the security dispensing handle **680** may comprise a grasp, knob, dial **686**, or other feature configured to assist a customer or user in moving the secure product dispenser assembly **600** between an open and closed position. In the embodiment shown, in the closed

position, the products **10a**, **16a** are prevented from being removed or added at the front **602** end of the tray **630**, while at the same time the products **10a**, **16a** may be removed and/or added at the back end **601** of the tray **630**. When the secure product dispenser is installed onto a display shelf, the back end **601** of the tray **630** is inaccessible and products are only able to be accessed from the front end **602** of the tray **630**. Installation of the secure product dispenser assembly **600** may comprise coupling or otherwise securing one or more components of the secure product dispenser assembly **600** to the display surface **604** or display shelf.

(63) Referring to FIG. **13**, the secure product dispenser assembly **600** is depicted in the open position. Rotation of the security dispensing handle **680** about an axis of rotation R (FIG. **13**) rotates the lip **684** such that the product **10a** can be removed from the dispensing position **636** of the front end **602** of the tray **630**. The rotation of the lip **684** thereby prevents removal of product **16a**, which is retained by the lip **684**, the top portion **670**, the tray **630**, and the first and second dividers **610**, **620**. In the open position, only the product positioned in the dispensing position **636** (FIG. **11**) may be removed from the secure product dispenser **600**. In an embodiment, the rotation of the security dispensing handle **680** may cause interaction between the plurality of surface features **683** and the one or more catch elements (not shown) in order to enable incremental movement of the security dispensing handle **680**. The interaction may also create an audible sound, such as a clicking noise, as the security dispensing handle **680** is rotated in order to draw attention and discourage tampering with the secure product dispenser assembly **600**.

(64) Rotation of the security dispensing handle **680** back to the closed position depicted in FIGS. **11-12** allows the next unit of product **16a** to move or slide along the tray **630** and into the dispensing position **636**. In the embodiments shown, the secure product dispenser assembly **600** is reloaded with product by decoupling from the display surface or display shelf in order to remove the secure product dispenser assembly **600** to expose the back end **601** of the tray **630**. Products may be loaded or unloaded from the back end **601** of the tray **630**. In other embodiments, the interior space of the secure product dispenser assembly **600** may not be accessible without removal of one or more components.

(65) In an embodiment, the top portion **670** may be configured to be adjusted relative to the first and second dividers or sides **610**, **620**. The embodiment illustrated in FIG. **14** provides height adjustability. The top portion **670** includes an adjustable height interface **677** that includes a plurality of holes **678** that can mate with a ball-pin **619** located on a divider or side interface. For example, to set the first (left) divider **610** at the lowest height, the adjustable height interface **677** would be inserted into the first divider interface **615** until the ball-pin **619** was inserted into the upper or highest hole **678** on the adjustable height interface **677**. On the other hand, to set the first (left) divider **610** at the highest height, the adjustable height interface **677** would be inserted into the first divider interface **615** until the ball-pin **619** was inserted into the lower or lowest hole **678** on the adjustable height interface **677**. In this embodiment, the second side or divider **620** may be adjusted in a similar manner at the second divider interface **625**.

(66) Referring to FIG. **13**, the secure product dispenser assembly **600** is depicted in the open position. Rotation of the security dispensing handle **680** about an axis of rotation R (FIG. **13**) rotates the lip **684** such that the product **10a** can be removed from the dispensing position **636** of the front end **602** of the tray **630**. The rotation of the lip **684** thereby prevents removal of product **16a**, which is retained by the lip **684**, the top portion **670**, the tray **630**, and the first and second dividers **610**, **620**. In the open position, only the product positioned in the dispensing position **636** (FIG. **11**) may be removed from the secure product dispenser **600**. In an embodiment, the rotation of the security dispensing handle **680** may cause interaction between the plurality of surface features **683** and the one or more catch elements (not shown) in order to enable incremental movement of the security dispensing handle **680**. The interaction may also create an audible sound, such as a clicking noise, as the security dispensing handle **680** is rotated in order to draw attention and discourage tampering with the secure product dispenser assembly **600**.

(67) Rotation of the security dispensing handle **680** back to the closed position depicted in FIGS. **11-12** allows the next unit of product **16a** to move or slide along the tray **630** and into the dispensing position **636**. In the embodiments shown, the secure product dispenser assembly **600** is reloaded with product by decoupling from the display surface or display shelf in order to remove the secure product dispenser assembly **600** to expose the back end **601** of the tray **630**. Products may be loaded or unloaded from the back end **601** of the tray **630**. In other embodiments, the interior space of the secure product dispenser assembly **600** may not be accessible without removal of one or more components.

(68) In an embodiment, the top portion **670** may be configured to be adjusted relative to the first and second dividers or sides **610**, **620**. The embodiment illustrated in FIG. **14** provides height adjustability. The top portion **670** includes an adjustable height interface **677** that includes a plurality of holes **678** that can mate with a ball-pin **619** located on a divider or side interface. For example, to set the first (left) divider **610** at the lowest height, the adjustable height interface **677** would be inserted into the first divider interface **615** until the ball-pin **619** was inserted into the upper or highest hole **678** on the adjustable height interface **677**. On the other hand, to set the first (left) divider **610** at the highest height, the adjustable height interface **677** would be inserted into the first divider interface **615** until the ball-pin **619** was inserted into the lower or lowest hole **678** on the adjustable height interface **677**. In this embodiment, the second side or divider **620** may be adjusted in a similar manner at the second divider interface **625**.

(69) As shown in FIGS. **15-17**, multiple secure product dispenser assemblies **600** may be arranged side-by-side on a display surface **604** or shelf in order to securely display and dispense a variety of products. Multiple secure product dispenser assemblies **600** are arranged side-by-side on a display shelf **604** such as would be used in a grocery store or pharmacy.

(70) In an embodiment, the bottom side or tray of any of the secure product dispenser assemblies discloses may comprise a first portion configured to move relative to a second portion to accommodate products a varying width. The configuration of such a bottom side or tray may be similar to any of the adjustable sides or dividers previously disclosed. In another embodiment, the bottom side may comprise a first portion and a second portion. The first portion may be configured to move relative to the second portion such that the first side or divider and second side or divider can be advanced towards or away from each other. In this manner, the secure product dispensing assembly is enabled to accommodate products of varying widths.

(71) Turning now to FIGS. **18-20**, an embodiment of the secure product dispenser **800** is shown comprised of a plurality of metal wires **803** or rails that form a wire frame defining a plurality of openings **807**. The wire frame of this embodiment of the secure product dispenser assembly **800** may form one or more elements that are similar to embodiments previously described, such as a base portion **805** and a top portion **870**. In an embodiment, at least a portion of the wire frame may be coated or otherwise treated with one or more substances to inhibit structural degradation from various chemical elements.

(72) The base portion **805** may comprise, in an embodiment, a first base portion **805a** (see FIG. **22**) and a second base portion **805b** (see FIG. **22**) that are moveable relative to each other to enable the base portion to be widened or narrowed to accommodate products of varying size. The base portion **805** further includes a first (left) divider **810**, a second (right) divider **820**, a back portion **840**, and a dispensing area **850** at least partially defined by a front stop member **851** at a front end **802** of the base portion **805**. The front stop member **851** may be configured or otherwise sized to allow a label, tag, or other indicator may be affixed or coupled thereto. An extension **860** or angular adjustment member may project from or be coupled to a bottom surface **806** of the base **805** to tilt the base **805** to enable gravitational forces to assist in moving the products **10a**, **16a-c** from the back end **801** of the base **805** to a front end **802** of the base **805** as products **10a**, **16a-c** are removed from the secured product dispenser assembly **800**. In another embodiment, a spring-loaded pusher is configured to advance additional units of product towards the dispensing area **850** to replace

units of product being dispensed to customers.

(73) As shown in FIGS. **18** and **22-25**, the extension **860** is configured to lift a back end or rear end **801** of the base **805** relative to the front end **802** or first end of the base **805** such that the base **805** is positioned at an angle β relative to a display surface **804**. The angles α (FIG. **9**) and β may be the same or may be different from each other. The base **805** may further comprise one or more friction reduction elements or slide elements on a base surface **808** configured to elevate the products **10a**, **16a-c** from the base surface **808** thereby decreasing the surface friction between the base surface **808** and the products **10a**, **16a-c**. The decreased surface friction enables the products **10a**, **16a-c** to slide more freely along or relative to the base surface **808** when being dispensed.

(74) The first and second dividers **810**, **820** may have similar features as the dividers of other embodiments of the secure product dispenser assembly previously described. In an embodiment, the first and second dividers **810**, **820** and the base **805** may be a single wire frame component. In another embodiment, the first and second dividers **810**, **820** may be configured to adjust in height in order to accommodate products of varying heights. The height of the first and second dividers **810**, **820** may be adjusted according to any of the embodiments previously discussed.

(75) As shown specifically in FIG. **19**, the top portion **870** is configured to extend between and couple to the first and second dividers **810**, **820** and an interior space is delimited by the base **805**, the top portion **870**, the first divider **810**, the second divider **820**, and the back portion **840**. In other embodiments, the top portion **870** may be similar to the lip **124** discussed previously discussed. Still referring to FIG. **19**, the top portion **870** extends from the back end **801** towards the front end **802** of the secure product dispenser **800** to prevent products **10a**, **16a-c** from being vertically removed from the top of the secure product dispenser assembly **800**. In an embodiment, the top portion **870** may be configured to be disengaged from the first and second dividers **810**, **820** in order to assist in the loading of product **10a**, **16a-c**. In an embodiment the top portion **870** may be configured to be locked into engagement with the first and second **810**, **820** dividers to prevent disengagement.

(76) Referring to FIGS. **19** and **23-27B**, the top portion **870** is coupled to a security dispenser handle assembly **880** toward the front end **802** of the secure product dispenser **800**. In an embodiment, the security dispensing handle assembly **880** is injection molded from a transparent or opaque resin and is configured to be fixedly coupled to the top portion **870**. As shown, the security dispensing handle assembly **880** comprises a body **882** with a top surface **881** that may be embossed with surface features or lettering **883**. A dispenser handle **886** is moveably coupled to the body **882** and a lip **884** may be operatively coupled to the dispenser handle **886**. It may be seen in FIG. **24** that an embodiment of the dispenser handle **886** can comprise a substantially circular shape and the lip **884** may comprise a shape that is compatible with the units of products **10a**, **16a-c**. The lip **884** may be comprised of transparent, translucent, or opaque plastic, or a metal.

(77) Turning to FIGS. **26A** and **27A**, the dispensing handle assembly **880** is depicted in the closed position. In the closed position, the top portion **870**, the first and second dividers **810**, **820**, the lip **884**, and the front stop **851** all act to prevent removal of the products **10a**, **16a-c** from the front end **802** of the secure product dispenser **800**. In an embodiment, one or more additional side guards **855** (FIG. **25**) may be provided around the dispensing area **850** to further retain a unit of product **10a** within the dispensing area while providing added protection against theft. As can be seen, when the secure product dispenser assembly is in the closed position, the lip **884** is configured to contact and otherwise aid in retaining the top portion of the product **10a**. Referring specifically to FIGS. **26A-B**, the dispenser handle **886** may comprise one or more grasps **887** or other feature configured to assist a customer or user in operating the dispenser handle **886** to move the secure product dispenser assembly **800** between an open and closed position. Installation of the secure product dispenser assembly **800** may comprise coupling or otherwise securing one or more components of the secure product dispenser assembly **800** to the display surface **804** or display shelf. In an embodiment, the dispenser assembly **800** may be secured to the display surface **804** using one or

more security screws **809** (FIGS. 21-22) positioned at various locations.

(78) The security dispensing handle assembly **880** may comprise a lock assembly **890** that enables the dispensing handle to be moved between a dispensing mode and a loading mode. In order to dispense units of product **10a**, **16a-c** from the dispenser assembly **800**, a lock member **891** of the lock assembly **890** is moved to or positioned in the “normal” or dispensing position (FIG. 26A). The customer may then grasp and rotate the dispenser handle **886** which acts to cause rotation of the lip **884** from the closed position as shown in FIG. 27A, to an open position shown in FIG. 27B where the lip **884** is positioned between the unit of product **10a** in the dispensing area **850** and the next unit of product **16a** set to enter the dispensing area **850**. The customer may then remove the unit of product **10a** from the dispensing area **850**. The customer may rotate the dispenser handle **886** back to the closed position or in an embodiment, the dispenser handle **886** may be capable of rotating back to the closed position automatically, such as due to the presence of a spring-loaded mechanism in the dispensing handle assembly **880**. The tilted nature of the base **805** enables gravitational force to aid movement of the next unit of product **16a** into the dispensing area **850**.

(79) In previously discussed embodiments, the secure product dispenser assembly may be reloaded from the rear end and/or from the top, however this embodiment of the secure product dispenser assembly **800** may be front loaded by moving the lock member **891** of the lock assembly **890** from the dispensing position (FIG. 26A) to the loading position (FIG. 26B). When in the loading position, the lock assembly **890** enables the dispensing handle assembly **880** to be held in a loading position such that the lip **884** is rotated out of the closed position and held to a left or right side of the dispensing area **850** in order to enable units of product to be loaded into the dispensing area **850** and pushed towards the rear end **801** of the secure product dispenser assembly **800**. Once loading is complete, the lock element **891** is moved back to the dispensing position, which causes the lip **884** to automatically return to the closed position. The dispensing handle assembly **880** is now ready to be operated by the customer to securely dispense units of product.

(80) As shown in FIG. 28, the secure product dispenser assembly **800** may be secured in a side-by-side configuration in a similar manner as previously discussed embodiments. However, multiple secure product dispenser assemblies **800** may be coupled together at their side surfaces by one or more coupling members **819**. As shown, in FIG. 29A, the one or more coupling members **819** may be in the form of one or more hooks configured to engage a side surface of an adjacent secure product assembly **800**. It is also conceivable that the secure product dispenser assembly **800** may be mounted to or otherwise hung from a wall as shown in FIG. 29B.

(81) In another embodiment shown in FIGS. 30A-B, the one or more coupling members **819** may be positioned on a top surface **870** of the secure product dispenser assembly **800**. In this embodiment, the coupling members **819** of each secure product dispenser assembly **800** may act to engage and couple adjacent secure product dispenser assemblies **800** together. The coupling members **819** may be formed as a single component with the top surface **870** or may be removeably coupled to the top surface **870** to increase the adjustability and the ability to customize. As shown, the coupling members **819** may be hooks configured to engage a portion of the wire frame of an adjacent secure product dispenser assembly **800**. Each secure product dispenser assembly **800** may then be coupled to a display surface **804** using any of the methods previously discussed. In this manner, adjacent secure product dispenser assemblies **800** are secured to the display surface **804** as a single “block”, thereby inhibiting vertical and horizontal movement relative to the display surface **804**.

(82) The components of the disclosed embodiments of the secure product dispensing assembly may be comprised of a rigid material resistant to fractures and moisture such as a durable plastic, a metal such as steel, or any combinations thereof.

(83) As described above, the present invention can be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope

of the invention to those having ordinary skill in the art. Furthermore, all “examples” or “example embodiments” given herein are intended to be non-limiting and among others supported by representations of the present invention.

PARTS LIST

(84) 10 first product **12** neck (product) **14** base (product) **16** second product **100** secure product dispenser assembly **110** first (left) divider **112** divider top portion **113** security dispensing handle mount **114** divider lip **115** divider bottom portion **116** divider back portion **117** divider adjustable height interface **118** holes **119** ball-pin **120** second (right) divider **130** tray **132** tray back portion **140** pusher **150** facing **200** security dispensing handle **202** security dispensing handle inner ring **204** security dispensing handle outer ring **210** gripping element **220** spring mechanism **230** gear mechanism **232** teeth rack **300** locking assembly **310** first locking rail **312** first locking rail slot **314** first locking rail mount **320** second locking rail **322** second locking rail teeth **324** second locking rail hook **330** hook **340** shelf **350** lock **400** secure product dispenser assembly **410** first (left) divider **414** divider lip **420** second (right) divider **424** divider lip **500** security dispensing handle **600** secure product dispenser assembly **601** back end **602** front end **604** display surface **610** first divider **615** first divider interface **619** ball-pin **620** second divider **625** second divider interface **630** tray **631** bottom surface, tray **632** top surface, tray **634** friction reducing element, tray **636** dispensing position **640** tray back portion **650** dispensing area **651** face **660** extension **670** cap **672** security dispenser handle coupler **677** adjustable height interface **678** plurality of holes **680** security dispenser handle **682** top surface, security dispenser handle **683** surface features, security dispenser handle **684** lip, security dispenser handle **700** secure product dispenser assembly **701** unit(s) of product **705** base portion **710** first side **715** first side interface **720** second side **725** second side interface **730** bottom side **740** back side **745** back side interface **750** dispensing area **752,753,754** dispensing area sides **770** top portion **771** first retainer **772** second retainer **774** first side interface engager **775** second side interface engager **776** back side interface engager **780** dispensing handle **781** first rotating portion **782** inner ring **783** catch portion **784** outer ring **785** second rotating portion **786** inner ring opening **790** stop member **792** stop member mount **794** rearward facing surface **796** forward facing surface **800** Secure product dispenser assembly **801** Back end **802** Front end **803** Wires **804** Display surface **805** Base portion **806** Bottom surface **807** Opening **808** Base surface **809** Security screw **810** First divider **819** Coupling member **820** Second divider **840** Back portion **850** Dispensing area **851** Front top member **855** Side guard **860** Extension **870** Top portion **880** Security dispenser handle assembly **882** Body **884** Lip **886** Dispenser handle **887** Grasp **890** Lock assembly **891** Lock member

Claims

1. A secure product dispenser comprising: a base extending along a base axis between a first end and an opposing second end, and configured to hold a plurality of units of product; a top portion configured to adjustably connect to the base; and a dispenser portion coupled to the top portion and positioned at least partially over the first end of the base, wherein the dispenser portion is configured to rotate between an open position and a closed position, wherein in the open position, the dispenser portion is configured to inhibit removal of more than one unit of product from the plurality of units of product held by the base by inhibiting movement of another unit of product into a position of the one unit product after the one unit of product is removed from the base, and wherein in the closed position, the dispenser portion is configured to inhibit removal of any of the plurality of units of product held by the base.
2. The secure product dispenser of claim 1, wherein the base comprises: a first side comprising a first side interface and an opposing second side comprising a second side interface.
3. The secure product dispenser of claim 2, wherein: the top portion further comprises first and second adjusting interfaces; the first adjusting interface adjustably couples to the first side interface

- and the second adjusting interface adjustably couples to the second side interface; and the first adjusting interface is configured to move relative to the first side interface and the second adjusting interface is configured to move relative to the second side interface in order to accommodate units of product of varying heights.
4. The secure product dispenser of claim 1, wherein the dispenser portion is configured to rotate about an axis of rotation, wherein the axis of rotation intersects the base axis.
 5. The secure product dispenser of claim 1, further comprising an angular adjustment member configured to elevate the second end of the base relative to the first end of the base such that the plurality of units of product are gravity-fed from the second end of the base to the first end of the base.
 6. The secure product dispenser of claim 1, further comprising a lock configured to hold the dispensing portion in a loading position, wherein when in the loading position, units of product can be loaded from the first end of the base.
 7. The secure product dispenser of claim 1, wherein rotation of the dispenser portion is configured to produce an audible indication.
 8. A method of manufacturing a secure product assembly, the method comprising: structuring a base to extend along a base axis between a first end and an opposing second end and to hold a plurality of units of a product; structuring a top portion to adjustably connect to the base; coupling a dispenser portion to the top portion; structuring the dispenser portion to rotate between an open position and a closed position, inhibiting removal of more than one unit of product from the plurality of units of product held by the base when the dispenser portion is in the open position by inhibiting movement of another unit of product into a position of the one unit product after the one unit of product is removed from the base; and inhibiting removal of any of the plurality of units of product from the base when the dispenser portion is in the closed position.
 9. The method of claim 8, further comprising structuring the base to comprise: a first side comprising a first side interface and an opposing second side comprising a second side interface.
 10. The method of claim 9, further comprising: structuring the top portion to comprise first and second adjusting interfaces; structuring the first adjusting interface to adjustably couple to the first side interface and structuring the second adjusting interface to adjustably couple to the second side interface; and moving the first adjusting interface relative to the first side interface and moving the second adjusting interface relative to the second side interface in order to accommodate units of product of varying heights.
 11. The method of claim 8, further comprising structuring the dispenser portion to rotate about an axis of rotation that intersects the base axis.
 12. The method of claim 8, further comprising structuring an angular adjustment member to: couple to the second end of the base, and to elevate the second end of the base relative to the first end of the base such that the plurality of units of product are gravity-fed from the second end of the base to the first end of the base.
 13. The method of claim 8, further comprising structuring the dispensing portion to define a loading position, wherein when in the loading position, units of product can be loaded from the first end of the base.
 14. The method of claim 13, further comprising structuring a lock to hold the dispensing portion in the loading position.
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